D 32727 (**Pages : 2**) **Name.....**

Reg. No.....

FIRST SEMESTER M.Sc. DEGREE (C.B.C.S.S.) [REGULAR /SUPPLEMENTARY] EXAMINATION, NOVEMBER 2022

Microbiology

MBG 1C 02—BIOPHYSICS AND INSTRUMENTATION

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

Part A (Short answer types)

Answer **four** *out of 6 questions.*

Answer in 2 **or** *3 sentences.*

1. Peptide bond.
2. SEM.
3. Beta sheets.
4. Redox potential.
5. CCD Camera.
6. TLC.

(4 × 2 = 8 weightage)

Part B (Short Essay types)

Answer **four** *out of 6 questions.*

7. Compare the working principle of confocal and polarization microscopy.
8. Differentiate AGE, PAGE and Native PAGE.
9. Compare IR spectra and Raman spectra.
10. Describe the laws of thermodynamics.
11. Differentiate HPLC and GLC.
12. Give details on DNA-Histone interaction.

(4 × 3 = 12 weightage)

Turn over

Part C (Essay types)

Answer **two** *out of 4 questions.*

13. Discuss about various physico-chemical forces and interactions.
14. Discuss on principle and applications of various types of centrifugations.
15. Give the principle and applications of Ultrafiltration, sonication and lyophilization.
16. Describe Ramachandran plot and give its importance.

(2 × 5 = 10 weightage)